

Different DC-DC Converters Recently Released/Announced by Different Manufacturers

Table 1

Model	Make	Application	Input voltage DC range	Output voltage DC range	Output current	Output power	Soft start	Digital interface	Efficiency	RoHS compliant	Safety/quality standards
BMR683	Flex	Industrial	36V to 60V	18V to 32V	17.9A at 500W	500W	Yes	Yes	95.50%	Yes	IEC/EN/UL 62368-1
PUB-2M series	Flex	Industrial	5V to 24V	5V to 12V	0.167 to 0.4A	2W	NA	No	86.50%	Yes	IEC/EN/UL 62368-1
BD9P1X5EFV-C	Rohm	Automotive	3.5V to 40V	0.8V to 8.5V	1A	NA	Yes	NA	NA	NA	FS supportive/AEC-Q100
BD9P2x5EFV-C Series	Rohm	Automotive	3.5V to 40V	0.8V to 8.5V	2A	NA	Yes	NA	NA	NA	FS supportive/AEC-Q100
BD71850MWV	Rohm	Industrial	2.7V to 5.5V	0.7V to 3.3V	1.5A to 3A	NA	Yes	Yes	83% to 95%	NA	NA
UWE-Q12 Series	Murata	Industrial	9V to 36V	5V to 24V	5A to 24A	120W	Yes	NA	90% to 92%	Yes	IEC/EN/UL/CSA
AP61100Q	Diodes Inc	Automotive	2.3V to 5.5V	0.6V to 3.6V	1A	NA	Yes	Yes	89.00%	Yes	AEC-Q100
R05CT05S	Recom	Industrial	4.5V to 5.5V	3.3V or 5V	100mA	500mW	NA	NA	60.00%	NA	NA
URD48xxxD-30WR3	Mornsun	Industrial	18V to 75V	5V to 24V	0.6A to 4A	30W	NA	NA	84.00%	Yes	CISPR32/EN55032/IEC61000-4-4/EC61000-4-5/EN62368
ORQB-50Y05x	Bel	Industrial	24V to 110V	5V	10A	50W	NA	NA	86 to 88%	Yes	IEC/EN/CSA/UL 62368-1
TMR 4 series	Traco Power	Industrial	9V to 75V	5V to 24V	166mA to 800mA	4W	NA	NA	82% to 87%	Yes	IEC/EN/UL 62368-1

NA indicates the parameters are not clearly mentioned in datasheet

ble operation mode through digital input that turns the converter on or off.

Power management feature is included in some newer DC-DC converters. For example, Flex's BMR683 DC-DC converter has power management feature that can be configured using its digital interface. Its software suite can be used to configure and monitor this product via the PMBus interface. It incorporates a wide range of configurable power management features that are simple to implement with a minimum of external components. Rohm's BD71850MWV DC-DC converter is a programmable power management device for powering single-core, dual-core, and quad-core SoCs such as NXP-i.MX 8M Nano.

Efficiency is an important design parameter of DC-DC converters. Along with innovation and introduction of CMOS devices, digital control can improve the efficiency of a previously analogue design. With careful design, modern switching regulators/converters can achieve 90 percent efficiency and even more. For example, BMR683 can have high efficiency, typically 95.5 percent at

full load. BD71850MWV has its efficiency target between 83 percent and 95 per cent. Murata's UWE-Q12 series DC-DC converters have efficiency up to 92 per cent.

Multi-input multi-output (MIMO) feature in newer DC-DC converters allows users to integrate multiple input sources and multiple output loads simultaneously. That is, this type of converter allows input sources to be added or removed seamlessly without cross-regulation problem. It is easy to achieve high extension capability for arbitrary inputs and outputs. For example, BD71850MWV integrates six buck regulators and six low-dropout (LDO) regulators to provide all the power rails required by the SoC and the commonly used peripherals.

The soft-start function in DC-DC converter is used to prevent large inrush current and output voltage overshoot. The function could be with an internally fixed soft-start time, or externally adjustable soft-start time, setting pin and externally adjustable oscillator duty cycle. Soft-start function is included in many DC-DC converters including BMR683, BD9P1X5EFV-C, BD9P2x-

5EFV-C, BD71850MWV, UWE-Q12 series, and AP61100Q.

Some DC-DC converters are integrated with remote control function, offering a convenient way to control your application. The control could be through a wire or wireless using infrared (IR) or some other wireless technology. For example, UWE-Q12 series has a remote sense circuit through which the product can be controlled using IR technology. TMR 4 series converter has dedicated pin for remote control function.

Ensuring safety is paramount to product users and industry. Adequate safety measures include fail-safe design against physical injury, damage to property, failure, or malfunction of a product. There are many industry safety standards that require restriction of hazardous substances or RoHS compliance, such as IEC/EN/UL safety requirements, Automotive Electronics Council (AEC) quality standards, etc. For example, BMR683, PUB-2M series, UWE-Q12 series, ORQB-50Y05x and TMR 4 series meet safety requirements according to IEC/EN/UL62368-1, while BD9P1X5EFV-C, BD9P2x5EFV-C, and AP61100Q are AEC-Q100 qualified. **EFY**